Lab 1

Antoni Smółka

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1. DICOM, NIfTI and BIDS

Each DICOM file holds one 2D slice plus acquisition metadata (3 T Siemens TrioTim). DICOM stores times in ms: TR = 1950 ms, TE = 2.26 ms.

Acquisition 4 vs 16. Acquisition 4 is a T1w MPRAGE: 3D, 1 mm isotropic, 176 slices, TR/TE 1950/2.26 ms. Acquisition 16 is DTI with 63 directions: 2D EPI, 2 mm isotropic, 64 MOSAIC volumes, TR/TE 9200/92 ms, plus b-values and b-vectors.

JSON with pydicom. `dicom2nifti` writes no `.json`, so I built the sidecar from the DICOM header (manufacturer, field strength, TR/TE in seconds, flip angle, slice thickness). `dcm2niix` writes it automatically, and `-f` controls the file name.

BIDS. I converted all 5 DICOM sessions with `dcm2niix` into `BIDS-mine/sub-CONxx/ses-*/{anat,dwi}`. `sub-CON01` has only a T1w scan. All 29 files pass `BIDSValidator().is_bids()`.

Missing files. Only `dataset_description.json` (with `Name` and `BIDSVersion`) is required. `README`, `CHANGES`, `participants.tsv` and `participants.json` are recommended. Every image also needs a JSON sidecar, and DWI needs `.bval`/`.bvec`.

2. Visualization and the affine

`sub-CON01` T1w: $176 \times 256 \times 256$ voxels, 1 mm, RAS orientation.

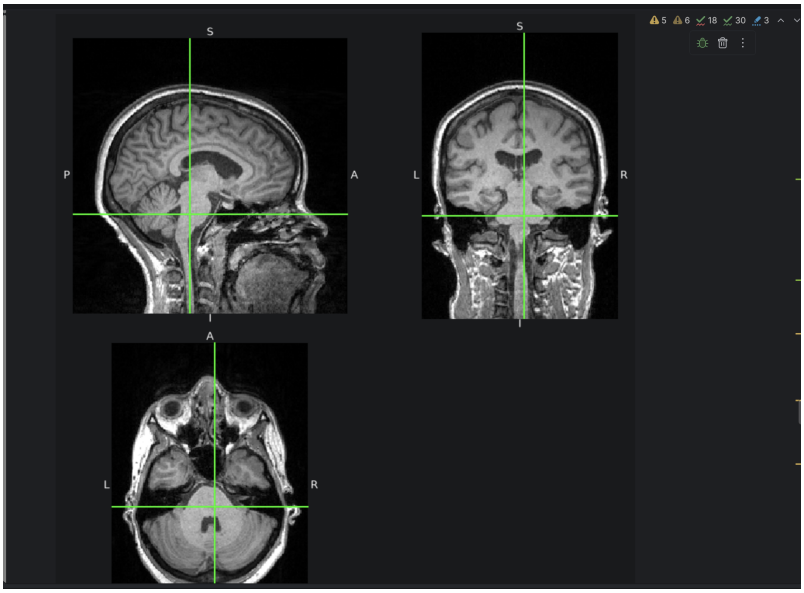


Figure 1: sub-CON01 T1w in three orthogonal planes.



Figure 2: sub-CON01 (red) over sub-CON02 in nilearn: the two scans do not overlap.

The two scans have different affines. The same voxel (60, 140, 140) is at (-26.2, 44.4, 40.8) mm in `sub-CON01` and at (-27.7, 58.2, 27.4) mm in `sub-CON02`. Each scan is in its own subject space, so the scans must be registered before they can be compared.

Why nilearn uses the NIfTI object. The object carries the affine (voxel → mm). Nilearn uses it to place, resample and overlay images. A plain numpy array has no spatial information.

3. FSL: brain extraction, segmentation, registration

FSL 6.0.7.23 was run on `sub-CON01`.

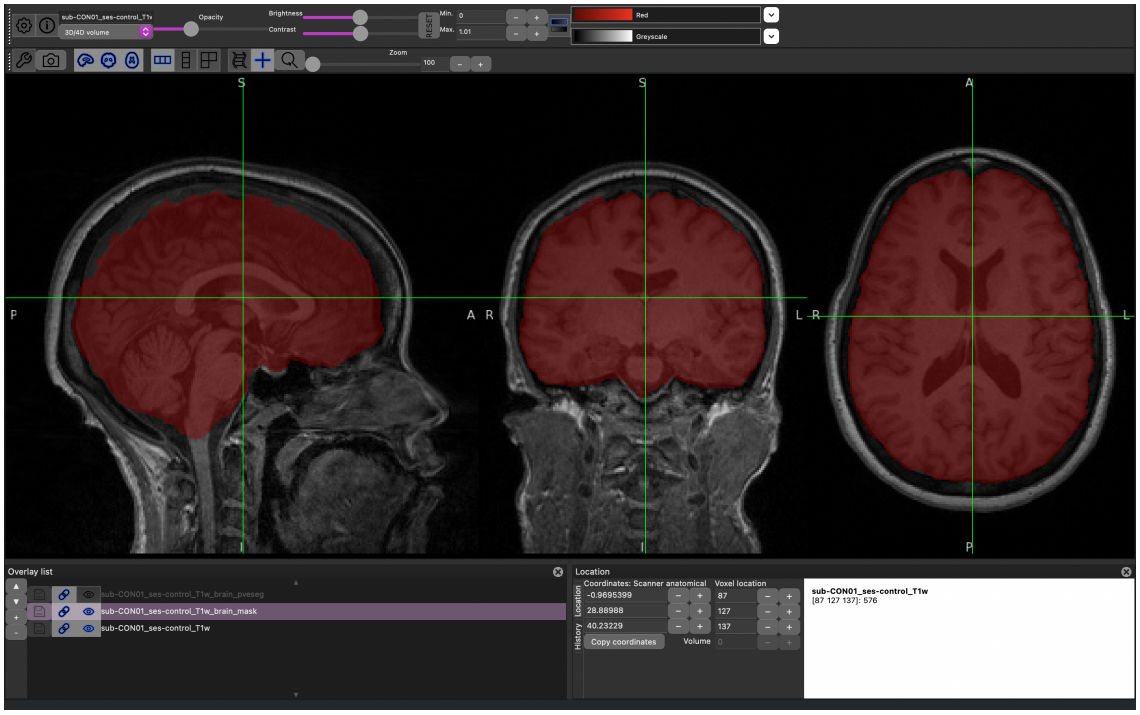


Figure 3: `bet` brain mask (red) on the T1w, two passes (`-f .5 -g -.6`, then `-f .35 -m`). A single default pass kept the neck and eyes, which broke the registration.

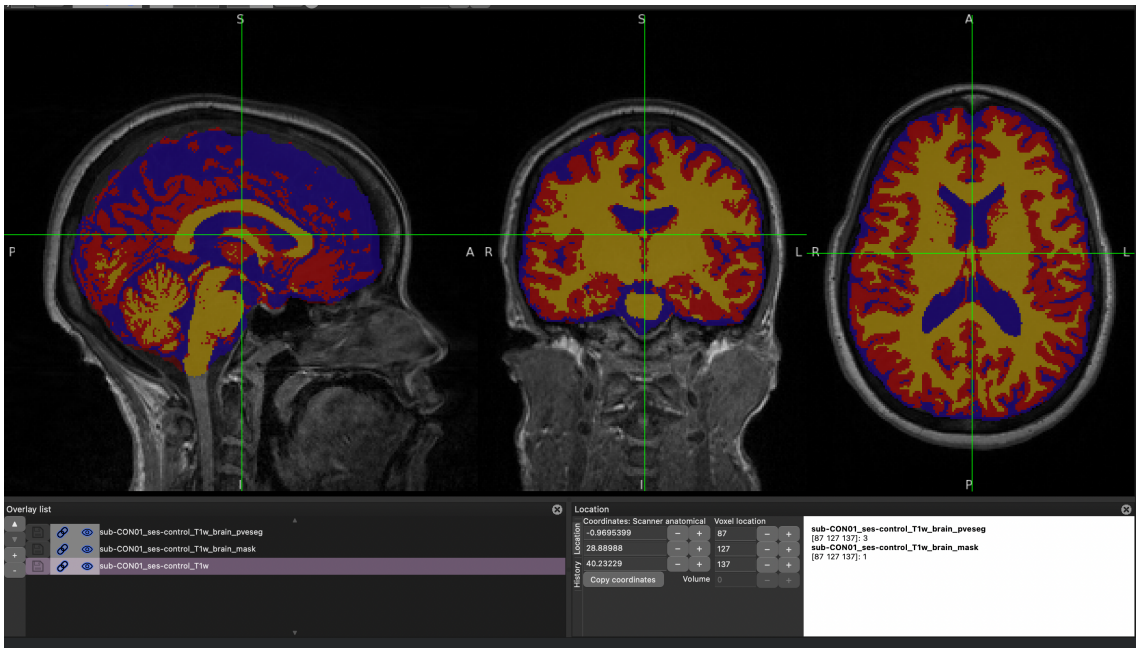


Figure 4: `fast` segmentation: CSF (blue) ≈ 341 ml, GM (red) ≈ 581 ml, WM (yellow) ≈ 559 ml.



Figure 5: `flirt` (12 DOF affine): the registered brain (hot, 50% opacity) over the MNI152 1 mm template.

All three methods are unsupervised. `bet` uses a deformable surface, `fast` uses a Gaussian mixture with an MRF and bias-field correction, and `flirt` optimizes an image-similarity cost (correlation ratio).